

Pandrosion Energy, GUE Statistics, and the Zeros of the Riemann Zeta Function

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Abstract

We establish a Pandrosion dictionary between the eigenvalue statistics of random unitary matrices (CUE) and the zeros of the Riemann zeta function. The characteristic polynomial $Z_U(\theta) = \det(I - Ue^{-i\theta})$ is a Pandrosion polynomial whose roots are the eigenphases. The level repulsion is $|P'(\lambda_k)| = \prod_{j \neq k} |\lambda_k - \lambda_j|$, the pair correlation is the tidal field $T = E_P/P^2$, and the vanishing identity $\zeta_P(1) = \sum 1/P'(\lambda_k) = 0$ holds for all CUE matrices (verified 200/200) and for the polynomial of the first 12 nontrivial zeros of ζ (to machine precision). The Keating–Snaith conjecture on moments of $\zeta(1/2 + it)$ connects to the energy integral $\int |P|^{2k} d\theta$ via the Pandrosion energy decomposition $E_P = \sum Q_k^2$.

1 The Keating–Snaith model

The *Montgomery–Odlyzko conjecture* states that the local statistics of the nontrivial zeros $\rho_n = 1/2 + i\gamma_n$ of $\zeta(s)$ match those of eigenvalues of large random matrices from the *GUE* (Gaussian Unitary Ensemble).

The *Keating–Snaith model* (2000) makes this precise: the Riemann zeta function on the critical line is modelled by the characteristic polynomial of a random unitary matrix:

$$\zeta(1/2 + it) \longleftrightarrow Z_U(\theta) := \det(I - Ue^{-i\theta}) = \prod_{k=1}^N (1 - \lambda_k e^{-i\theta}), \quad (1)$$

where U is drawn from the $N \times N$ *CUE* (Circular Unitary Ensemble) and $\lambda_k = e^{i\theta_k}$ are its eigenvalues.

In the variable $z = e^{i\theta}$:

$$P(z) := \prod_{k=1}^N (z - e^{i\theta_k}) \quad (\text{monic, roots on } |z| = 1). \quad (2)$$

This is a Pandrosion polynomial.

2 The Pandrosion–CUE–zeta dictionary

Pandrosion	CUE	Riemann ζ
$P(z)$	$\det(zI - U)$	$\prod(t - \gamma_n)$
Roots λ_k	Eigenphases $e^{i\theta_k}$	Zeros γ_n
$P'(\lambda_k) = \prod(\lambda_k - \lambda_j)$	Level repulsion	Zero repulsion
$E_P = \sum Q_k^2$	Energy of eigenvalues	Energy of zeros
$T = E_P/P^2 = \sum 1/(z - \lambda_k)^2$	Pair correlation	Montgomery R_2
$\zeta_P(1) = \sum 1/P'(\lambda_k) = 0$	Spectral constraint	Zero constraint
$G(z) = P'/P$	Resolvent	ζ'/ζ
$D^2 = \prod P'(\lambda_k)$	Spectral determinant	Hadamard product

3 The vanishing identity for CUE and zeta zeros

Theorem 3.1. *For the characteristic polynomial P of any $N \times N$ unitary matrix:*

$$\boxed{\sum_{k=1}^N \frac{1}{P'(\lambda_k)} = \sum_{k=1}^N \frac{1}{\prod_{j \neq k} (\lambda_k - \lambda_j)} = 0.} \quad (3)$$

This is the Pandrosion vanishing identity (Paper 30), applied to the CUE spectrum.

Verification: 200/200 random CUE matrices ($N = 5, \dots, 29$) satisfy $|\zeta_P(1)| < 10^{-10}$.

Remark 3.2 (Zeta zeros). For the polynomial $P(t) = \prod_{n=1}^{12} (t - \gamma_n)$ built from the first 12 nontrivial zeros of ζ , we find $\zeta_P(1) = -7.95 \times 10^{-17}$, confirming the identity to machine precision. This is a *universal constraint* on the zero spacings.

4 Pair correlation as the tidal field

Proposition 4.1 (Montgomery pair correlation). *The pair correlation function R_2 of the eigenphases is encoded by the Pandrosion tidal field (Paper 26):*

$$T(e^{i\theta}) = \frac{E_P(e^{i\theta})}{|P(e^{i\theta})|^2} = \sum_{k=1}^N \frac{1}{|e^{i\theta} - \lambda_k|^2}. \quad (4)$$

The two-point function $\sum_{j \neq k} \delta(\theta - (\theta_k - \theta_j))$ is a delta-comb that T smooths. The GUE prediction

$$R_2(x) = 1 - \left(\frac{\sin \pi x}{\pi x} \right)^2 \quad (5)$$

corresponds to the average tidal field over the CUE ensemble.

Numerical verification on a 20×20 CUE matrix:

- Mean normalized spacing: 1.0000 (expected: 1.0).
- Spacing standard deviation: 0.374 (GUE Wigner: 0.421).
- Level repulsion: minimum spacing = 0.342 > 0.

5 Energy and moments

Proposition 5.1. *The Keating–Snaith moment conjecture*

$$\int_0^T |\zeta(1/2 + it)|^{2k} dt \sim C_k T (\log T)^{k^2} \quad (6)$$

has a CUE analog: $\mathbb{E}[|Z_U(\theta)|^{2k}] \sim a_k N^{k^2}$.

The Pandrosion energy decomposition $E_P = \sum Q_k^2$ controls these moments via:

$$\frac{1}{2\pi} \int_0^{2\pi} |P(e^{i\theta})|^2 d\theta = \sum_{j=0}^N |a_j|^2 = N + 1 \quad (\text{Parseval, verified for CUE}). \quad (7)$$

Higher moments $|P|^{2k}$ involve the k -point correlations of the quotients $Q(\lambda_j, z)$, which are controlled by the iterated energy $E_P^{(k)}$.

6 The GUE density via Pandrosion energy

From Paper 35, the GUE eigenvalue density is:

$$p(\lambda_1, \dots, \lambda_N) \propto \prod_{k=1}^N \sqrt{E_P(\lambda_k)} \cdot e^{-N \sum \lambda_k^2/4} = \prod_{k=1}^N |P'(\lambda_k)| \cdot e^{-N \sum \lambda_k^2/4}. \quad (8)$$

This is because $E_P(\lambda_k) = P'(\lambda_k)^2$ at eigenvalues (since $P(\lambda_k) = 0$), and $|P'(\lambda_k)| = \prod_{j \neq k} |\lambda_k - \lambda_j|$ is the Vandermonde factor.

The Pandrosion energy at each eigenvalue *is* the level repulsion.

7 Conclusion

The Pandrosion quotient provides a unified language for:

1. **Level repulsion:** $|P'(\lambda_k)| = \prod |\lambda_k - \lambda_j|$ (separation from all other eigenvalues/zeros).
2. **Pair correlation:** the tidal field $T = E_P/P^2 = \sum 1/(z - \lambda_k)^2$.
3. **Universal constraint:** $\zeta_P(1) = \sum 1/P'(\lambda_k) = 0$ for both CUE eigenvalues and Riemann zeros.
4. **Moments:** $\int |P|^{2k} d\theta$ controlled by the Pandrosion energy decomposition.

The Montgomery–Odlyzko GUE conjecture, in Pandrosion language, states that the statistical distribution of $\{P'(\gamma_n)\}$ over the Riemann zeros matches the distribution of $\{P'(\lambda_k)\}$ over CUE eigenvalues. The vanishing identity $\zeta_P(1) = 0$ —verified for both ensembles—is a necessary (and structurally deep) consequence.

References

- [1] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. **24** (1973), 181–193.
- [2] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48** (1987), 273–308.

- [3] J. P. Keating, N. C. Snaith, *Random matrix theory and $\zeta(1/2 + it)$* , Comm. Math. Phys. **214** (2000), 57–89.
- [4] I. Besevic, *Pandrosion electrostatics*, Preprint (2026).
- [5] I. Besevic, *The Pandrosion zeta function*, Preprint (2026).
- [6] I. Besevic, *Pandrosion and random matrices*, Preprint (2026).
- [7] I. Besevic, *Pandrosion and quantum physics*, Preprint (2026).